

CAPE NEWENHAM, AK, USA

WMO: 703050

Lat: 58.650N

Lon: 162.067W

Elev: 165

StdP: 99.36

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25623

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -22.7	(c) -20.8	(d) -26.1	(e) 0.4	(f) -22.4	(g) -24.0	(h) 0.4	(i) -20.2	(j) 19.7	(k) -5.6	(l) 18.2	(m) -5.1	(n) 5.4	(o) 10	(p) 0.913

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 3.0	(c) 16.9	(d) 13.0	(e) 15.3	(f) 12.4	(g) 14.1	(h) 11.8	(i) 14.0	(j) 15.6	(k) 13.1	(l) 14.6	(m) 12.4	(n) 13.8	(o) 3.5	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 13.0	(b) 9.5	(c) 14.4	(d) 12.4	(e) 9.1	(f) 13.8	(g) 11.7	(h) 8.7	(i) 13.2	(j) 39.4	(k) 15.7	(l) 37.3	(m) 14.7	(n) 35.6	(o) 13.9	(p) 18.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	15.7	14.1	-23.2	19.2	2.6	2.6	-25.0	21.1	-26.6	22.6	-28.0	24.1	-29.9	26.0
			-23.1	15.1	2.1	1.4	-24.6	16.1	-25.9	16.9	-27.1	17.7	-28.6	18.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-6.6	-4.7	-5.6	-0.8	4.1	8.3	10.8	11.2	8.8	3.5	-1.1	-4.7
	DBStd	8.06	7.82	6.73	6.39	4.43	3.39	2.67	2.32	1.96	2.33	3.74	5.17	6.84
	HDD10.0	3045	513	411	483	324	183	64	18	9	49	201	333	456
	HDD18.3	5972	772	644	742	574	440	301	234	223	287	459	583	714
	CDD10.0	115	0	0	0	0	2	14	42	45	12	1	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.1	6.0	7.3	6.6	6.1	5.1	4.1	4.5	5.7	6.2	6.8	7.5	7.3
Precipitation	PrecAvg	830	66	62	51	42	48	51	70	81	105	93	88	74
	PrecMax	1219	152	162	142	102	92	89	114	129	199	184	175	128
	PrecMin	636	19	7	10	9	13	8	30	20	13	16	13	15
	PrecStd	135	29	32	33	20	20	18	22	27	41	39	42	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	5.1	5.2	9.0	15.3	18.5	19.3	18.0	15.5	11.7	9.0	6.7
		MCWB	3.8	3.6	3.2	4.9	9.3	13.2	14.9	14.2	12.8	9.7	7.5	5.3
	2%	DB	4.6	4.1	3.8	6.9	12.4	15.7	17.2	16.2	13.5	10.3	7.4	5.0
		MCWB	3.1	2.7	2.1	4.1	8.1	11.4	13.5	13.4	11.5	8.7	6.0	3.8
	5%	DB	3.5	3.2	2.8	5.4	10.2	13.9	15.2	14.9	12.6	9.2	6.2	3.9
		MCWB	2.2	1.9	1.3	3.2	7.1	10.3	12.6	13.0	10.7	7.8	4.9	2.7
	10%	DB	2.5	2.5	1.9	4.1	8.8	12.3	14.0	13.9	11.7	8.1	5.2	2.9
		MCWB	1.4	1.4	0.7	2.5	6.4	9.6	11.9	12.4	10.1	6.8	4.2	1.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.4	4.0	3.8	5.7	10.1	13.9	15.8	15.1	13.4	10.1	7.7	5.7
		MCDB	5.8	4.8	4.7	7.9	13.9	16.8	18.7	16.8	14.5	11.0	8.6	6.5
	2%	WB	3.3	3.1	2.4	4.4	8.5	12.2	14.0	14.0	11.9	8.9	6.3	4.0
		MCDB	4.3	3.8	3.5	6.4	11.5	15.1	16.0	15.3	13.1	10.1	7.3	4.9
	5%	WB	2.5	2.2	1.6	3.5	7.5	10.9	13.0	13.2	11.1	8.0	5.3	2.9
		MCDB	3.4	3.0	2.6	5.1	10.0	13.1	14.8	14.4	12.2	9.0	6.2	3.7
	10%	WB	1.5	1.5	0.8	2.7	6.5	9.9	12.3	12.6	10.3	7.1	4.3	1.9
		MCDB	2.4	2.3	1.8	4.1	8.6	11.8	13.7	13.7	11.4	8.1	5.3	2.8
Mean Daily Temperature Range	5% DB	MDBR	4.8	4.9	4.7	4.4	4.3	4.6	3.6	3.0	2.8	3.2	4.0	4.6
		MCDBR	4.1	3.8	4.4	5.2	6.7	6.8	5.6	4.5	3.5	3.0	3.5	4.2
		MCWBR	3.2	3.0	3.3	3.4	4.2	4.0	3.4	2.9	2.7	2.8	3.2	3.5
	5% WB	MCDBR	3.9	3.8	4.1	4.8	6.3	6.3	4.8	4.1	3.1	2.8	3.4	3.7
		MCWBR	3.3	3.4	3.4	3.3	4.1	4.1	3.2	2.8	2.7	2.8	3.4	3.4
Clear-Sky Solar Irradiance	taub	0.260	0.305	0.304	0.360	0.382	0.365	0.378	0.373	0.344	0.313	0.293	0.260	
	taud	2.158	2.173	2.237	2.182	2.280	2.430	2.429	2.444	2.473	2.400	2.139	2.048	
	Ebn at Noon	563	705	837	840	847	871	849	827	796	707	494	405	
	Edh at Noon	47	77	96	122	119	104	102	94	78	62	48	37	
All-Sky Solar Radiation	RadAvg	0.41	1.06	2.64	4.07	5.08	5.20	4.34	3.47	2.44	1.27	0.47	0.22	
	RadStd	0.08	0.20	0.45	0.45	0.43	0.39	0.46	0.39	0.30	0.12	0.06	0.03	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page